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TSYP 14

IEEE TUNISIAN STUDENT AND YOUNG PROFESSIONAL CONGRESS

IAS TUNISIA SECTION CHAPTER
IEEE INDUSTRY APPLICATIONS SOCIETY

IAS CHALLENGE

TSYP14 TECHNICAL CHALLENGE

SCOPE & TOPIC

ENGINEERING INDUSTRY FOR A DYNAMIC FUTURE

Industrial systems are designed to operate efficiently under defined conditions. However, real-world industrial environments continuously evolve due to:

- Changing demands
- Unexpected disruptions
- Resource constraints
- Operational challenges
- Technological advances

The challenge focuses on developing innovative engineering solutions addressing real industrial needs, with emphasis on:

Resilience • Efficiency • Sustainability • Digitalisation

Challenge problem

Teams start from an industrial problem or process, preferably from a real industrial environment.

The solution must follow a research- and process-based methodology:

Industrial Problem → Process Mapping → Research & State of the Art → Solution Exploration → Comparative Analysis → Solution Development → Validation → Impact Assessment

The industrial case may come from a real industrial environment or from an organiser-provided / organiser-validated industrial case.

Challenge goal

Develop innovative, practical, research-driven and technically sound solutions that address real industrial challenges and prepare industrial systems for a more dynamic and demanding future.

Three Focus Areas

Resilient Industrial Systems

Efficient & Sustainable Industrial Systems

Digitalised Industrial Systems

All three follow the same fundamental approach:

Industrial Problem → Process Mapping → Research & State of the Art → Solution Development → Demonstration → Impact Assessment

Challenge Methodology

1. Industrial Problem

Identify and clearly characterize a relevant industrial problem.

2. Process Mapping

Map and analyse the existing AS-IS process, including information flows, resources, constraints and operational bottlenecks.

3. Research & State of the Art

Investigate relevant scientific research, standards, patents, modern industrial practices and existing solutions.

4. Solution Exploration

Explore possible engineering solutions.

5. Comparative Analysis

Compare the proposed alternatives and justify the selected approach.

6. Solution Development

Develop the engineering solution and demonstrate its feasibility.

7. Validation

Evaluate the technical solution and its performance.

8. Impact Assessment

Assess quantitative impact, including relevant technical, economic and environmental metrics.

Challenge Phases

Phase 1: Selection

Technical Poster

Teams submit a technical poster focused on:

- Relevance of the industrial problem
- AS-IS process mapping
- Literature and state-of-the-art research
- Validation of the proposed methodology

Phase 2: Development

Technical Report & Proof of Concept

Teams develop their solution through:

- 5–7 page expanded technical report
- Implementation of the Proof of Concept (PoC)
- Technical prototype / simulation / demonstration
- Technical presentation

The report and PoC should demonstrate the progression from the initial AS-IS process toward the proposed TO-BE process.

Phase 1: Selection

MVP & IAS Session at TSYP14

Teams must demonstrate:

- A mature and functional Minimum Viable Product (MVP)
- Quantitative impact assessment
- Technical, economic and environmental evaluation
- Final presentation and pitch

Delivrerables

Phase 1

- Technical Poster

Phase 2

- 5–7 page Technical Report + Proof of Concept

Phase 3

- Minimum Viable Product + Final Pitch
- Supporting materials may include:
  - Presentation
  - GitHub Repository
  - Hardware or software prototype
  - Report
  - Demo
  - PoC
  - Pitching video

Deliverables follow a progressive approach according to each phase's requirements.

Team Requirements

Maximum team size: **4 members**

Phase 2: Final phase

Finalists must provide:

- Pitching & Q&A
- Complete physical prototype
- Proposed solution
- Technologies used
- Project architecture
- User manual
- GitHub Repository

Pitching

English | 5 min pitching | 5 min Q&A

Scoring

TOTAL SCORE: 100 POINTS

Phase 1 — Selection: 25 pts

Technical Poster

- Problem Identification & AS-IS Mapping: 12 pts
  - Clarity and characterization
  - Industrial relevance
  - Accuracy of AS-IS process mapping
  - Information flows, resources, constraints and operational bottlenecks
- Research,State of the Art & Solution Exploration:13 pts
  - Scientific literature
  - Standards
  - Patents
  - Modern industrial practices
  - Exploration of 2–3 alternative approaches
  - Comparative methodology

Phase 2 — Development: 45 pts

Technical Report & PoC

- Technical Report Rigor & Methodology: 15 pts
- Proof of Concept (PoC) Implementation: 15 pts
- Technical Presentation: 15 pts

Phase 3 — Final: 30 pts

MVP & Pitching

- MVP Functionality & Industrial Impact Assessment: 20 pts
- Final TSYP14 Presentation & Pitch — 10 pts

Evaluation Focus

Phase 1

Problem → AS-IS Mapping → Research → State of the Art

Phase 2

Methodology → Solution → PoC → Technical Presentation

Phase 3

MVP → Impact → Validation → Final Pitch

The evaluation emphasizes:

- Industrial relevance
- Research quality
- Technical feasibility
- Innovation
- Engineering methodology
- Implementation maturity
- Quantitative impact
- Technical, economic and environmental performance

Important dates

- Challenge Infosession: **13/09/2026**
- Phase 1 Submission Deadline: **30/10/2026**
- Final Submission Deadline: **12/12/2026**

Note: The challenge proposal states that the Phase 2 submission and announcement dates may still change depending on the final Phase 2 presentation date.

Winners

3 WINNING TEAMS

- **1st Place:** 300 TND
- **2nd Place:** Certificate
- **3rd Place:** Certificate

Contact

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Submission

Submit your project and show us what you've built.

Contact



Please contact us through the following email address ias.tn@ieee.org for any inquires or clarifications.